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FORC studies of exchange biased NiFe in $L1_0$(111) FePt-based spin valve

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Abstract.
We use First-Order Reversal Curves (FORC) to study the switching distribution and exchange bias in $L1_0$ (111) FePt-based spin valves with a layer structure of Ta (6)/Pt (3)/$L1_0$ (111) $Fe_{53}Pt_{47}$ (20)/$Co_{50}Fe_{50}$ (1.5)/Cu (4.5)/$Co_{50}Fe_{50}$ (2)/$Ni_{80}Fe_{20}$ (3) (in nm). We find that the switching of the free layer magnetization is strongly influence by the magnetic state of the FePt/CoFe fixed layer, as evidenced by tunable coercivity and exchange bias fields.

1. Introduction

Exchange bias continues to be the topic of intense research due to its fundamentally interesting basic properties [1] and its application in spintronic devices, such as in magnetic read heads and magnetoresistive random access memories [2]. While originally observed in antiferromagnetic/ferromagnetic (AFM/FM) systems, this phenomenon may also arise between two FM layers in spin valves with hard-FM/NM/soft-FM layers (NM: non magnetic) structure, both with in-plane anisotropy [3,4,5] or with perpendicular anisotropy [6]. Recently, we have reported pseudo spin valves (hard-FM/NM/soft-FM layers) with *tilted* anisotropy [7]. Spin valves with tilted fixed layer magnetization are fundamentally interesting in that they allow for the study of GMR over a wide range of relative angles in-plane and out-of-plane angles between the two FM layers. They also have the potential for improved performance in spin torque devices, such as spin torque MRAM [8] and spin torque oscillators (STO) [9].

In spin valve sensors or spin valve-based spintronic devices, such as STOs, it is desirable that the soft layer switches freely without any pinning effect from the fixed layer. Since tilted spin valves are a new kind of device, it is also fundamentally interesting to study the exchange coupling between the free and fixed layers in our pseudo spin valves with titled anisotropy. In this paper, the magnetic reversal behaviour of exchange-biased CoFe/NiFe free layer in $L1_0$ FePt-based spin valves is studied by the first-order reversal curve (FORC) method [10]. It is found that the exchange bias and coercivities of CoFe/NiFe free layer strongly depends on the magnetization state of the FePt/CoFe fixed layer and can be used to tune both its coercivity and its exchange bias field.

2. Experiments

The $L1_0$ (111) FePt-based spin valves have the general layer structure of *underlayer*/$L1_0$(111) $Fe_{53}Pt_{47}$ (20)/$Co_{50}Fe_{50}$ (1.5)/Cu (4.5)/$Co_{50}Fe_{50}$ (2)/$Ni_{80}Fe_{20}$ (3) (in nm) and were deposited on thermally oxidized silicon substrates using a magnetron sputtering system (ATC Orion-8) with a base pressure better than 5×10^{-8} Torr. The $Fe_{53}Pt_{47}$ (FePt) alloy films were fabricated by co-sputtering Fe and Pt from elemental targets at a nominal substrate temperature of 700C and subsequently *in-situ* annealed at the same temperature for 10 min. Ta (6)/Pt (3) underlayers were used to promote the (111)-preferred texture of the $L1_0$ FePt layer, to ensure a high coercivity, and to improve the surface roughness [7]. Permalloy ($Ni_{80}Fe_{20}$,NiFe) free layers were sputtered at room temperature using a stoichiometric target. Thin $Co_{50}Fe_{50}$ (CoFe) layers were deposited at room temperature at both the bottom (1.5 nm) and the top (2 nm) Cu interfaces to enhance the interface spin polarization. Finally, a 5-nm-thick Ta capping layer was deposited at room temperature. Magnetic properties of the spin valve were measured by an alternating gradient magnetometer (AGM). A first order reversal curve (FORC) technique [10] was employed to study the details of the magnetization reversal. After saturation, then decreasing the field to a reversal field H_R, and measuring the magnetization M as the applied field H_a was increased back to saturation, tracing out a FORC. A family of FORCs was measured at different H_R, with equal field spacing, thus filling the interior of the major hysteresis loop. The FORC distribution was defined as a mixed second order derivative:

$$\rho \quad = \quad -\frac{1}{2}\frac{\partial^2 M(H_R, H_a)}{\partial H_R, H_a} \tag{1}$$

The FORC diagram is plotted against (H_C, H_U) coordinates on a contour map, where H_C is the local coercive field and H_U is the bias field, accomplished by a simple rotation of the coordinate system defined by: H_U=(H_a+H_R)/2 and H_C=(H_a-H_R)/2. The FORCinel software [11] was employed for the processing and analysis of the FORC diagrams.

3. Results and Discussion

Fig. 1. displays the in-plane full hysteresis loop of the spin valve. The rapid switching of the NiFe-based free layer is well separated from the much more gradual switching of the FePt-based fixed layer. To further elucidate the switching behavior, we carry out a full set of FORCs and study whether the free layer switching is affected by the state of the fixed layer. We first apply a large magnetic field to set the fixed layer into a particular state. Here we focus on fully saturated (14 kOe), half-switched (-4.6 kOe), and fully saturated at negative field (-14 kOe). We then measure families of FORCs around the NiFe switching region, i.e. we set the saturation field to only 2 kOe and limit the FORC range so as to not disturb the magnetic state of the fixed layer. While it is obviously impossible to entirely avoid affecting the fixed layer state during the FORC measurement, we find that we are able to extract meaningful data from a careful compromise of the FORC fields.

The corresponding FORC diagrams are shown in Fig. 2. When the hard fixed layer is positively saturated (Fig. 2(a)), the FORC diagram has a distinct peak located at a bias field H_U=25 Oe, and with a coercivity field H_C=220 Oe, corresponding to the exchange bias field and the coercivity of the CoFe/NiFe free layer [12]. When we progressively switch the hard layer, the peak in the FORC diagram gradually moves. When the FePt layer is half switched (Fig. 2(b)), the peak is centered at H_U=0 and coercivity H_C=320 Oe. The net magnetization of the hard FePt/CoFe layer is zero in this case, which averages away any exchange bias but also leads to a distinctly increased coercivity. The 45% increase in H_C is likely due to a state of maximum interface frustration in the interaction between the hard layer and the free layer [13]. In other words, the demagnetized state of the FePt/CoFe layer likely consists of many

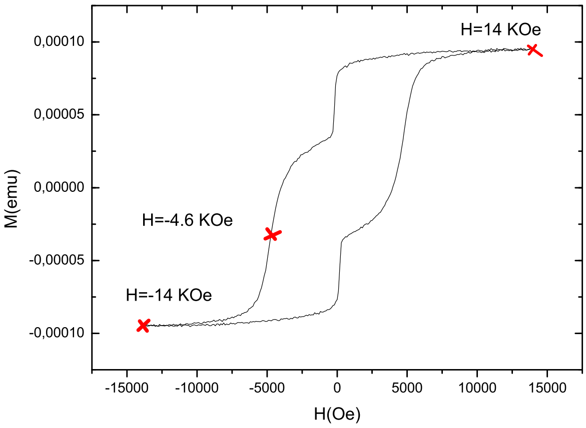

Figure 1. Full loop of $L1_0$ (111) FePt-based spin valve of Ta (6)/Pt (3)/$L1_0$ (111) FePt (20)/CoFe (1.5)/Cu (4.5)/CoFe (2)/NiFe (3).

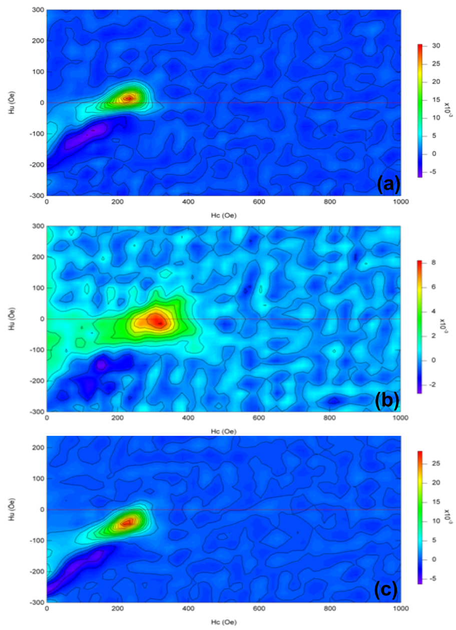

Figure 2. The FORC diagrams of (a) FePt/CoFe fixed layer is positively saturated;(b) FePt/CoFe fixed layer is half swithed;(c) FePt/CoFe fixed layer is negatively saturated.

oppositely arranged magnetic domains with local points of large associated stray field which can act has pinning centers during the switching of the free layer. When the hard layer is negatively saturated (Fig. 2(c)), the FORC diagram peak is shifted to negative with exchange bias H_U=-25 Oe, and the coercivity again returns to its lower value H_C=220 Oe. The two saturated states are hence even in coercivity and odd in the exchange bias field.

4. Conclusion

In summary, the exchange-biased CoFe/NiFe free layer in $L1_0(111)$ FePt-based spin valve is studied in detail using a FORC method. The exchange bias and coercivity of CoFe/NiFe free layer both depend on the magnetization state of the FePt/CoFe fixed layer. At FePt saturation, the same coercivity and the same maximum absolute exchange bias value is achieved at both negative and positive saturation field. However, when the FePt is prepared in a demagnetized state, the free layer coercivity peaks at a 45% greater value, and the exchange bias field goes to zero. We argue that both effects are due to a frustrated interface state between the free and the fixed layer as the fixed layer breaks up into a large number of small domains in its demagnetized state.

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